$^{34}S(n,\gamma)$ E=thermal 1985Ra15

1985Ra15: 6×10^{11} n/cm^2/s thermal neutrons at the target position were produced from the Los Alamos Omega West Reactor at a nominal 8-MW reactor power. γ rays following thermal neutron capture by a 1.09-g, 77.7% enriched ^{34}S target were detected using a 26 cm^3 coaxial Ge(Li) detector inside a 20-cm-diam by 30-cm-long NaI(Tl) annulus with FWHM=2.3 keV at 1 MeV, 5.5 keV at 6 MeV and 8.8 keV at 11 MeV. Measured σ_γ, Eγ, and Iγ for 61 γ-ray transitions. Deduced 22 levels and γ-ray branching ratios.

1972Dz13: Thermal neutrons were produced from the thermal column of the IRT reactor of the Baghdad Nuclear Research Institute. γ rays following thermal neutron capture by a 1.5-g, 97%-enriched ^{34}S target were detected using a 10-cm^3 Ge(Li) detector with FWHM=3.5 keV at 1 MeV and 8 keV at 7 MeV. Measured Eγ and Iγ for 22 γ-ray transitions. Deduced 11 levels.

1985Ke08: 5×10^{12} n/cm^2/s thermal neutrons were produced from the McMaster University Reactor. γ rays following thermal neutron capture by a natural sulfur target were detected using a *10%* efficient Ge centered within a NaI(Tl) annulus with FWHM=3.2-4.4 keV at 4-7 MeV. Measured Eγ and Iγ for 13 γ-ray transitions. Deduced 9 levels.

1997Be42: Thermal neutrons were provided from the reactor BR1 of the Studiecentrum voor kernenergie in Mol, Belgium. γ rays following thermal neutron capture by a 313.6-mg, 93.26%-enriched ^{34}S and 5.933%-enriched ^{36}S target were detected using a Ge detector. Measured σ_γ, Eγ, and Iγ for 11 γ-ray transitions. Deduced 8 levels. Compared with direct-capture (DC) model calculations.

All 55 Eγ values from 2007ChZX: database of prompt gamma activation analysis (PGAA) from neutron capture are in agreement with the data below.

^{35}S Levels

E(level)†	Comments
0	
1572.373 *7*	
1991.28 *4*	
2347.782 *8*	C^{2}S=0.48 for 3/2$^-$ from 1997Be42.
2716.991 *21*	
2938.63 *5*	
3558.081 *27*	
3801.954 *15*	C^{2}S=0.08 for 3/2$^-$ from 1997Be42.
4105.6 *8*	
4189.242 *33*	C^{2}S=0.15 for 1/2$^-$ from 1997Be42.
4477.60 *7*	
4903.374 *15*	C^{2}S=0.49 for 1/2$^-$ from 1997Be42.
4963.082 *25*	C^{2}S=0.19 for 3/2$^-$ from 1997Be42.
5752.5 *8*	
6018.8 *6*	
6078.476 *32*	
6293.93 *6*	
6354.89 *23*	
6419.9 *11*	
6629.43 *9*	
6761.0 *12*	
6986.098 *24*	

† From a least-squares fit to γ-ray energies.

$\gamma(^{35}S)$

Photons per 100 thermal neutron captures are reported in 1972Dz13 and 1985Ke08. γ-ray cross sections in mb are reported in 1985Ra15, multiplying these by 0.340 yields the number of photons per 100 thermal neutron captures. σ(4638)=163 mb *15*. Iγ(4638)=σ(4638)*0.340=55 *5*.

γ-ray intensities per incident neutron for all transitions and γ-ray cross sections in mb for primary transitions are reported in 1997Be42. The strongest 4638γ is used for normalization. σ(4638)=148.6 mb *55* and Iγ(4638)=σ(1997Be42)/σ(1985Ra15)*Iγ(1985Ra15)=50 *7*.

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^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ[†]	I_γ[†#]	E_i(level)	E_f	Comments
356.66[@] *9*	0.037[@] *4*	2347.782	1991.28	
356.66[@] *9*	0.037[@] *4*	6986.098	6629.43	
368.5 *4*	0.020 *7*	2716.991	2347.782	
619.23 *19*	0.042 *8*	3558.081	2938.63	
631.32[@] *24*	0.060[@] *9*	4189.242	3558.081	
631.32[@] *24*	0.060[@] *9*	6986.098	6354.89	
[x]646.9[‡] *5*	0.7 *1*			E_γ,I_γ: from 1972Dz13. 1972Dz13 assigned this γ ray as the 6986−>6339 transition. This γ ray and its final level 6339 have not been observed in any other ^{34}S(n,γ) measurements. The evaluators consider this γ ray questionable.
[x]663.41[‡] *7*	0.09 *1*			
692.16 *5*	0.133 *17*	6986.098	6293.93	
775.398 *6*	17.2 *19*	2347.782	1572.373	E_γ: other: 775.50 *15* (1972Dz13). I_γ: weighted average of 19.5 *20* (1972Dz13) and 15.6 *17* (1985Ra15).
[x]803.81[‡] *9*	0.092 *14*			
863.28 *28*	0.034 *10*	3801.954	2938.63	
907.608[‡] *23*	0.56 *10*	6986.098	6078.476	E_γ: weighted average of 908.1 *3* (1972Dz13) and 907.607 *14* (1985Ra15). I_γ: weighted average of 0.5 *1* (1972Dz13) and 0.61 *10* (1985Ra15).
1084.79 *15*	0.054 *10*	3801.954	2716.991	
1101.92 *31*	0.033 *8*	4903.374	3801.954	
1144.591 *20*	0.55 *6*	2716.991	1572.373	
1161.05 *20*	0.055 *8*	4963.082	3801.954	
1210.28 *4*	0.252 *27*	3558.081	2347.782	
1250.61 *5*	0.214 *24*	4189.242	2938.63	
[x]1381.67[‡] *24*	0.024 *8*			
1404.967 *24*	0.60 *6*	4963.082	3558.081	
1454.09 *4*	0.45 *11*	3801.954	2347.782	
1566.7 *3*	0.47 *9*	3558.081	1991.28	
1572.333 *8*	35.0 *34*	1572.373	0	E_γ: weighted average of 1572.15 *16* (1972Dz13) and 1572.333 *7* (1985Ra15). I_γ: others: 39 *4* (1972Dz13) and 32 *4* (1997Be42).
1760.55 *11*	0.146 *21*	4477.60	2716.991	
1840.2 *12*	2.2 *4*	4189.242	2347.782	E_γ: unweighted average of 1839 *1* (1972Dz13) and 1841.426 *15* (1985Ra15). I_γ: weighted average of 3.6 *8* (1972Dz13) and 2.14 *21* (1985Ra15).
1964.8 *2*	0.37 *10*	4903.374	2938.63	
1991.27 *5*	0.54 *6*	1991.28	0	
2022.954 *9*	11 *1*	6986.098	4963.082	E_γ: others: 2022.80 *20* (1972Dz13) and 2023.0 *10* (1985Ke08). I_γ: weighted average of 12.0 *15* (1972Dz13), 11 *1* (1985Ke08), 11.4 *10* (1985Ra15), and 9.1 *12* (1997Be42).
2082.83 *17*	15.9 *15*	6986.098	4903.374	E_γ: unweighted average of 2082.65 *16* (1972Dz13), 2083.17 *7* (1985Ke08), and 2082.681 *12* (1985Ra15). I_γ: weighted average of 18.0 *15* (1972Dz13), 17 *2* (1985Ke08), 15.6 *17* (1985Ra15), and 12.6 *17* (1997Be42).
2184.16[‡&] *19*	4.7 *5*	4903.374	2716.991	E_γ,I_γ: from 1985Ke08. 1985Ke08 assigned its final level at 2719, with the closest known level being 2717.0. This γ ray has not been observed in any other ^{34}S(n,γ) measurements. Including this γ ray in the level scheme would result in a significant intensity imbalance at its initial level 4903 and its final level 2717 and also lead to poor fitting in the level scheme. Evaluators consider this γ ray questionable.
2229.510 *16*	2.86 *27*	3801.954	1572.373	
2347.700 *15*	49 *4*	2347.782	0	E_γ: weighted average of 2347.50 *10* (1972Dz13), 2347.71 *3* (1985Ke08), and 2347.701 *11* (1985Ra15).

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$^{34}S(n,\gamma)$ E=thermal 1985Ra15 (continued)

$\gamma(^{35}S)$ (continued)

E_γ[†]	I_γ[†#]	E_i(level)	E_f	Comments
				I_γ: weighted average of 54 *5* (1972Dz13), 49 *4* (1985Ke08), 50 *5* (1985Ra15), and 40 *6* (1997Be42).
2508.39 *8*	0.38 *4*	6986.098	4477.60	
2555.492 *14*	3.14 *31*	4903.374	2347.782	E_γ: other: 2555.9 *4* (1972Dz13). I_γ: weighted average of 3.4 *7* (1972Dz13) and 3.09 *31* (1985Ra15).
2615.2 *2*	1.09 *10*	4963.082	2347.782	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2616.8 *13*	0.24 *7*	4189.242	1572.373	E_γ: other: 2614.3 *3* doublet (1972Dz13). I_γ: other: <1.3 *4* (1972Dz13).
2716.99 *16*	0.30 *5*	2716.991	0	
2796.74 *4*	4.9 *4*	6986.098	4189.242	E_γ: weighted average of 2796.83 *13* (1985Ke08) and 2796.73 *4* (1985Ra15). Other: 2796.6 *5* (1972Dz13). I_γ: weighted average of 6.1 *8* (1972Dz13), 4.3 *4* (1985Ke08), 5.4 *5* (1985Ra15), and 4.9 *7* (1997Be42).
2905.1 *4*	0.14 *4*	4477.60	1572.373	
2938.58 *11*	0.89 *9*	2938.63	0	
2972.0 *4*	0.18 *6*	4963.082	1991.28	
3139.9 *5*	0.078 *24*	6078.476	2938.63	
3183.95 *4*	6.1 *6*	6986.098	3801.954	E_γ: weighted average of 3184.30 *20* (1972Dz13), 3183.94 *7* (1985Ke08), and 3183.94 *4* (1985Ra15). I_γ: weighted average of 7.0 *9* (1972Dz13), 5.7 *6* (1985Ke08), 6.2 *6* (1985Ra15), and 5.7 *8* (1997Be42).
3330.84 *13*	7.6 *7*	4903.374	1572.373	E_γ: unweighted average of 3331.08 *15* (1972Dz13), 3330.64 *6* (1985Ke08), and 3330.80 *4* (1985Ra15). I_γ: weighted average of 7.8 *10* (1972Dz13), 7.7 *8* (1985Ke08), and 7.4 *7* (1985Ra15).
3390.62 *7*	5.6 *5*	4963.082	1572.373	E_γ: weighted average of 3390.86 *20* (1972Dz13), 3390.73 *7* (1985Ke08), and 3390.55 *5* (1985Ra15). I_γ: weighted average of 5.5 *8* (1972Dz13), 5.7 *6* (1985Ke08), and 5.5 *5* (1985Ra15).
3558.1 *5*	0.085 *17*	3558.081	0	
3801.73 *4*	2.59 *23*	3801.954	0	E_γ: weighted average of 3802.0 *3* (1972Dz13), 3801.49 *14* (1985Ke08), and 3801.74 *3* (1985Ra15). I_γ: weighted average of 3.6 *7* (1972Dz13), 2.3 *2* (1985Ke08), 3.09 *31* (1985Ra15), and 2.6 *4* (1997Be42).
4105.3 *8*	0.048 *17*	4105.6	0	
4188.95 *5*	2.54 *27*	4189.242	0	E_γ: weighted average of 4188.80 *13* (1985Ke08) and 4188.96 *4* (1985Ra15). Other: 4189.6 *3* (1972Dz13). I_γ: weighted average of 1.9 *4* (1972Dz13), 2.72 *27* (1985Ra15), and 2.8 *4* (1997Be42). Other: 24 *2* (1985Ke08).
4268.65 *14*	0.34 *4*	6986.098	2716.991	E_γ: other: 4268.4 *7* (1972Dz13). I_γ: other: 0.4 *2* (1972Dz13).
4638.14 *14*	56 *5*	6986.098	2347.782	E_γ: unweighted average of 4638.4 *2* (1972Dz13), 4638.10 *3* (1985Ke08), and 4637.91 *4* (1985Ra15). I_γ: weighted average of 56 *6* (1972Dz13), 59 *5* (1985Ke08), 55 *5* (1985Ra15), and 50 *7* (1997Be42).
4902.99 *5*	4.0 *4*	4903.374	0	E_γ: weighted average of 4903.07 *7* (1985Ke08) and 4902.96 *4* (1985Ra15). Other: 4903.4 *3* (1972Dz13). I_γ: weighted average of 3.2 *8* (1972Dz13), 4.3 *4* (1985Ke08), 3.7 *4* (1985Ra15), and 4.2 *6* (1997Be42).
4963.06 *22*	2.62 *27*	4963.082	0	E_γ: unweighted average of 4963.2 *5* (1972Dz13), 4963.34 *10* (1985Ke08), and 4962.63 *5* (1985Ra15). I_γ: weighted average of 2.6 *8* (1972Dz13), 2.92 *27* (1985Ra15), and 2.18 *33* (1997Be42). Other: 29 *3* (1985Ke08).
5752.0 *8*	0.018 *5*	5752.5	0	

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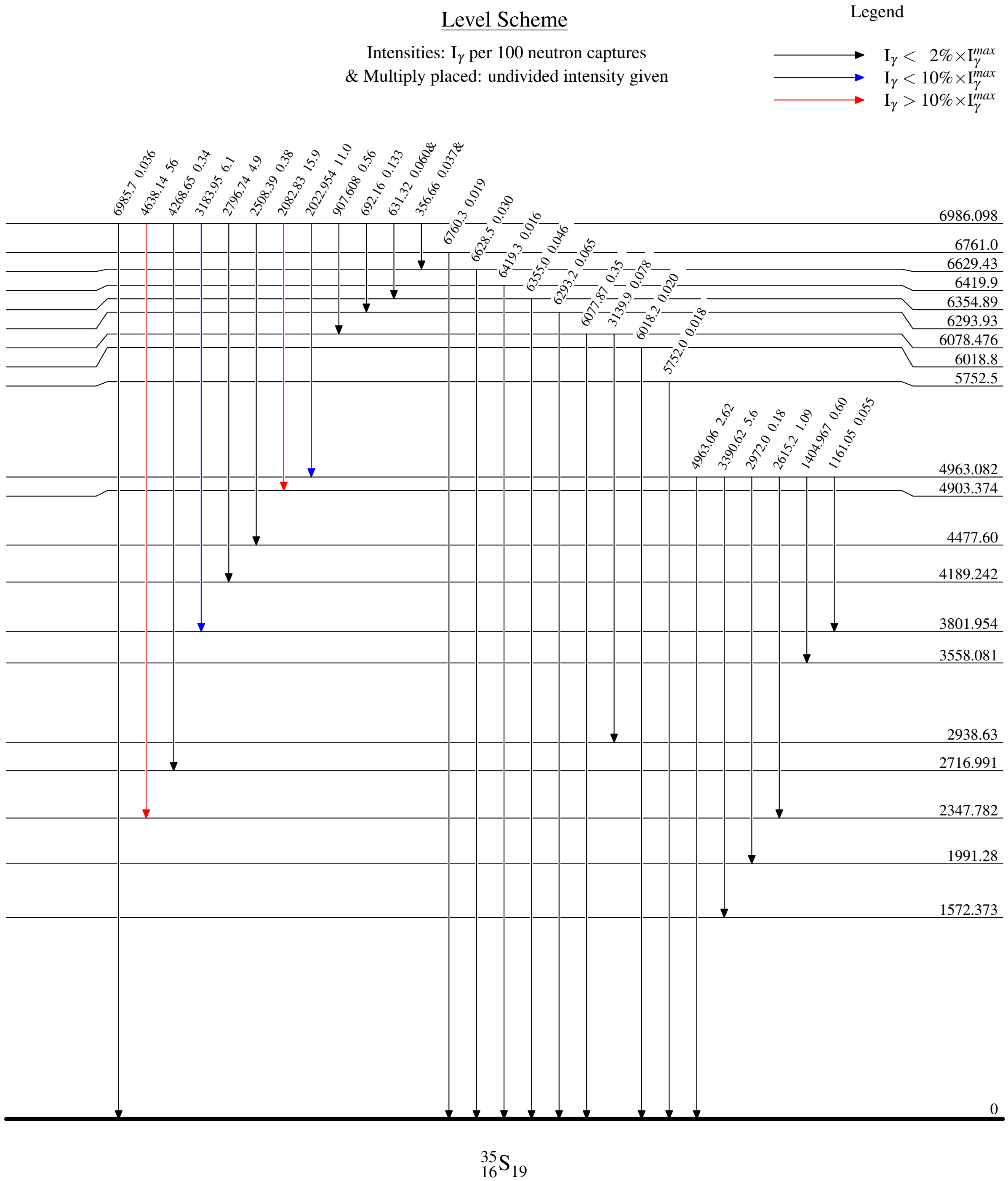
^{34}S(n,γ) E=thermal 1985Ra15 (continued)

γ(^{35}S) (continued)

E_γ †	I_γ †#	E_i(level)	E_f	Comments
6018.2 *6*	0.020 *7*	6018.8	0	
6077.87 *11*	0.35 *4*	6078.476	0	E_γ: other: 6078.2 *10* (1972Dz13). I_γ: weighted average of 0.3 *1* (1972Dz13) and 0.36 *4* (1985Ra15).
6293.2 *4*	0.065 *14*	6293.93	0	
6355.0 *6*	0.046 *7*	6354.89	0	
6419.3 *11*	0.016 *5*	6419.9	0	
6628.5 *6*	0.030 *6*	6629.43	0	
6760.3 *12*	0.019 *8*	6761.0	0	
6985.7 *10*	0.036 *8*	6986.098	0	

† From 1985Ra15, unless otherwise noted.
‡ Not observed in 2007ChZX (PGAA).
# Intensity per 100 neutron captures.
@ Multiply placed with undivided intensity.
& Placement of transition in the level scheme is uncertain.
x γ ray not placed in level scheme.

$^{34}S(n,\gamma)$ E=thermal 1985Ra15



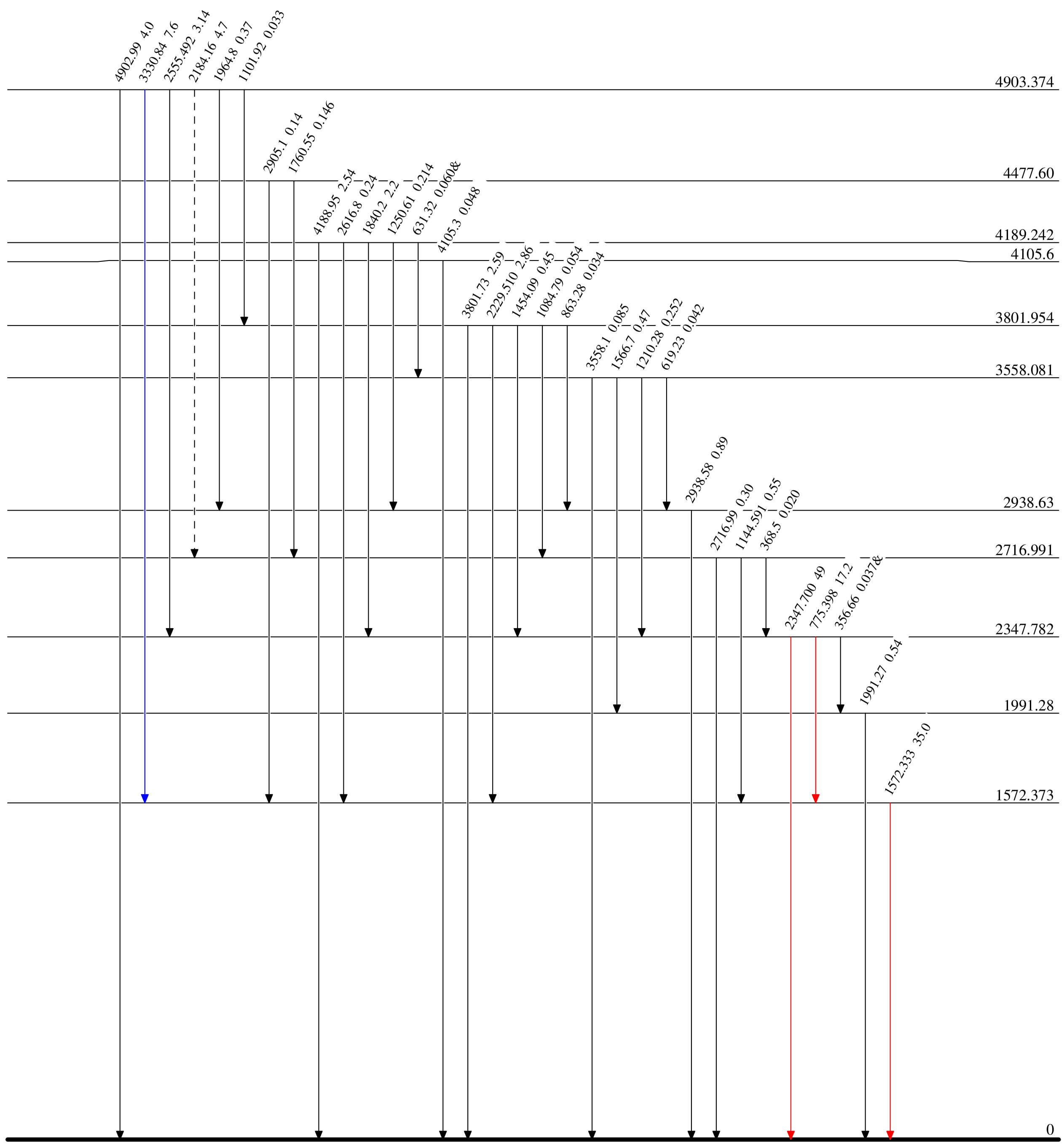
$^{34}S(n,\gamma)$ E=thermal 1985Ra15

Level Scheme (continued)

Intensities: I_γ per 100 neutron captures
& Multiply placed: undivided intensity given

Legend

$I_\gamma < 2\% \times I_\gamma^{max}$
$I_\gamma < 10\% \times I_\gamma^{max}$
$I_\gamma > 10\% \times I_\gamma^{max}$
γ Decay (Uncertain)



$^{35}_{16}S_{19}$